Devdoots Assignment 03 - Python OOPs (Encapsulation & Abstraction)

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1. Prerequisite Study Guide (is Assignment ke Liye Kya Padhna Hai?)

Is assignment ko attempt karne se pehle sabhi team members ko niche diye gaye topics aur concepts ko achhe se samajhna aur revise karna compulsory hai:

- **Encapsulation (Data Hiding & Protection):**
  - Public vs Private attributes in Python (Double underscore __ ka use).
  - Data integrity maintaining: Direct variable access ko roknas aur Getter/Setter methods ke zariye safe validation rules lagana.
  - Python's name mangling mechanism (kaise _ClassName__variable internal lock banata hai).
- **Abstraction (Complexity Hiding & Blueprint Enforcement):**
  - Abstract Base Class (ABC) ka concept: from abc import ABC, abstractmethod.
  - Blueprint Contract enforce karna: Parent class mein @abstractmethod define karna taaki har Child Class mandatory functions implement kare.
  - Abstract classes ka direct object na ban paana aur AI/ML libraries (jaise Scikit-Learn, PyTorch) mein iska real-world architecture implementation.

2. Assignment Questions & Tasks

Task 1: Secure Banking System (Encapsulation Practice)

Requirement: Ek SecureBankAccount class banayein jo balance aur PIN ko secure rakhe.

- Private Attributes: __balance (default 5000), __pin (default 1234).
- Public Attribute: account_holder.
- Methods:
  - withdraw(self, entered_pin, amount): PIN check kare, agar PIN sahi hai aur amount balance se kam/equal hai toh withdraw kare, warna error

message return kare.
    - change_pin(self, old_pin, new_pin): Sahi old PIN check karke new PIN update kare (New PIN 4-digit hona chahiye).

Task 2: AI Model Pipeline Blueprint (Abstraction Practice)

Requirement: Ek Abstract Base Class BaseAIModel(ABC) banayein.

- Abstract Methods (Decorated with @abstractmethod):
    - train_model(self, dataset_name)
    - evaluate_accuracy(self)
- Child Class 1: VisionModel(BaseAIModel) — In dono methods ko implement karke Image dataset par training aur accuracy string return kare.
- Child Class 2: NLPModel(BaseAIModel) — In dono methods ko implement karke Text dataset par training aur sentiment classification accuracy return kare.

3. GitHub Submission Rules & Workflow (Single File Push)

Main branch par direct commit karna strict मना hai. GitHub repository mein apni **single assignment file** push karne ke liye niche diye gaye workflow aur Git commands ko step-by-step follow karein:

Step	Git Command	Description
1. Sync Main Branch	`git checkout main` `git pull origin main`	Remote repository se latest code sync karein taaki local main branch updated rahe.
2. Create Feature Branch	`git checkout -b user/yourname-assign3`	Apni alag feature branch banayein (e.g., user/divyansh-assign3).
3. Create Assignment File	File Name: divyansh_assign3.py	Apne folder ke andar sirf apni specific assignment

Step	Git Command	Description
		file banayein aur code likhein.
4. Stage & Commit Single File	git add divyansh_assign3.py git commit -m "[HW-3] Completed OOPs Encapsulation and Abstraction" git push origin user/yourname-assign3	**Important:** git add . ke bajaye **sirf apni single file** (git add divyansh_assign3.py) stage karein aur push karein.
5. Raise Pull Request (PR)	GitHub Repo → Compare & pull request	GitHub portal par jaakar PR raise karein. Code review ke baad use main mein merge kiya jayega.